- In trees, vertices are usually called nodes;
- In trees edges are somehow implicit: since they can only go from a node to its children, it's more common to talk about parent/children relations than explicitly list edges. Also, because of this, trees are implicitly represented with adjacency lists.

Furthermore, trees have other peculiar characteristics that makes them a strict subset of the whole set of graphs; in particular, any tree is a *simple*, *undirected*, *connected* and *acyclic* graph.

We have illustrated in the previous section what *simple graph* means: a tree, in fact, can neither have multiple edges between two nodes, nor it can have loops; instead, only a single edge between a node and each of its children is allowed.

Let's now see what the other three properties mean.

14.2.1 Undirected

As we have mentioned in section 1, a graph is directed when all its edges can be traversed in a single direction, from source (the first vertex in the edge's pair) to destination.

In undirected graphs, conversely, the edges can be traversed in both directions.

An undirected graph can easily be represented as a directed graph, by expanding each undirected edge (u, v) into a couple of directed edges (u, v) and (v, u).